

# Design and Validation of a Gravity-Gradient Boom Arm for Hybrid Attitude Control of a PocketQube Satellite

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Attitude control systems are a critical component of satellite operations, enabling precise orientation and mission execution. For PocketQube satellites, implementing compact methods for attitude control systems reduces size, weight, and power consumption. Continued miniaturization enables broader access to space based sensing, communication, and technology demonstration missions. The Wentworth PocketQube is an experimental Earth imaging platform with a hybrid Attitude Control and Determination System platform. Magnetic torque coils provide two axes of instantaneous control within Earth's dipolar magnetic field, while a passive gravity gradient mast enables full three-axis stabilization. To ensure resilience to disturbance, the gravity gradient mast, a tethered mass, is suspended below the satellite. Utilizing the difference of the gravitational force acting on the mass and satellite, a stable and robust attitude is achieved. The following paper investigates the major design constraints, simulations, and experimentation performed before validation of flight readiness. This paper demonstrates that advanced attitude determination and control capabilities are achievable in PocketQube satellites using hybrid attitude control architectures. These capabilities are enabled through simulation-driven design and experimental validation, in this case to support high quality horizon imaging.

## I. Nomenclature

|                     |   |                                                                                                  |
|---------------------|---|--------------------------------------------------------------------------------------------------|
| $\mathbf{a}$        | = | Gravitational acceleration vector (m/s <sup>2</sup> )                                            |
| $C_{\text{deploy}}$ | = | Tether damping coefficient during deployment phase (N·s/m)                                       |
| $C_{\text{final}}$  | = | Tether damping coefficient at full deployment / friction end-stop (N·s/m)                        |
| $\mathbf{h}$        | = | Orbital angular momentum vector, $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ (m <sup>2</sup> /s) |
| $k$                 | = | Tether elastic stiffness coefficient (N/m)                                                       |
| $L$                 | = | Instantaneous tether length between attachment point and boom (m)                                |
| $\dot{L}$           | = | Rate of change of tether length (m/s)                                                            |
| $L_0$               | = | Initial tether length at deployment (m)                                                          |
| $L_{\text{max}}$    | = | Maximum tether length at deployment (m)                                                          |
| $L_{\text{spool}}$  | = | Length of tether paid out by spool (m)                                                           |

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|                  |                                                                                          |
|------------------|------------------------------------------------------------------------------------------|
| $n$              | = Mean orbital motion, $n = \sqrt{\mu/a^3}$ (rad/s)                                      |
| $q$              | = Attitude quaternion representing satellite orientation                                 |
| $\dot{q}$        | = Quaternion rate of change                                                              |
| $r$              | = Inertial position vector (m)                                                           |
| $r^2$            | = Squared orbital radius (m <sup>2</sup> )                                               |
| $T$              | = Tether tension force (N)                                                               |
| $v$              | = Inertial velocity vector (m/s)                                                         |
| $x_L$            | = Radial (nadir) axis of LVLH frame, $x_L = -r/\ r\ $                                    |
| $y_L$            | = Along-track axis of LVLH frame, $y_L = z_L \times x_L$                                 |
| $z_L$            | = Orbit-normal axis of LVLH frame, $z_L = (r \times v)/\ r \times v\ $                   |
| $\mu$            | = Earth's gravitational parameter, $3.986 \times 10^{14}$ m <sup>3</sup> /s <sup>2</sup> |
| $\omega$         | = Body angular velocity vector (rad/s)                                                   |
| $\omega_{LVLH}$  | = Angular velocity of the LVLH frame, $\omega_{LVLH} = h/r^2$ (rad/s)                    |
| $\Omega(\omega)$ | = Quaternion rate matrix                                                                 |

## II. Introduction

PocketQube satellites represent a rapidly evolving class of miniaturized spacecraft within modern satellite technology. A PocketQube is a standardized picosatellite platform, typically based on a cubic unit measuring 5 cm x 5 cm x 5 cm, with a total spacecraft mass generally below 1 kg [1]. Despite their compact form factor, PocketQubes provide a robust and adaptable platform for space-based experimentation. They are increasingly utilized by academic institutions, defense organizations, and commercial entities for technology demonstration, scientific research, and specialized mission applications [2]. The severe constraints on mass, volume, and power inherent to this platform drive innovative engineering solutions and enable rapid, cost-effective testing of emerging space technologies.

Attitude control presents a significant challenge for spacecraft of this scale. Larger satellites commonly employ thrusters, reaction wheels, or control rods to achieve attitude control. These methods are impractical for PocketQubes due to strict limitations on mass, power consumption, and internal volume [3]. Consequently, alternative approaches must be considered. The Wentworth Institute of Technology's LeopardSat-1 implements a hybrid attitude control architecture combining gravity gradient stabilization with magnetic torque coils, leveraging both Earth's gravitational field and magnetic field to achieve passive and semi-active control. Gravity gradient stabilization enables a specific face of the satellite to align toward Earth's center over multiple orbital periods.

Gravity gradient stabilization has been demonstrated in multiple early spacecraft missions. An initial implementation was deployed during the 1966 Gemini 11 missions, establishing the feasibility of gravity gradient concepts in orbit, although full stabilization was not achieved. The Department of Defense Gravity Experiment (DODGE) successfully demonstrated gravity gradient stabilization in 1967, followed by additional applications such as Explorer 49, which utilized gravity-gradient stabilization in lunar orbit for radio astronomical measurements [4]. These missions established gravity gradient stabilization as a viable passive attitude control strategy and provided foundational insight for adapting the technology to modern small satellite platforms.

For this mission profile, gravity gradient stabilization is achieved using a deployable boom arm consisting of a tethered mass extended beneath the satellite toward nadir. The boom is attached to the camera board located near the satellite's center and deploys from the bottom face of the satellite, which houses a camera and light sensors. The tether attachment point is intentionally offset from the spacecraft center of mass such that, once deployed, the extended mass dominates the system inertia and induces a restoring torque. This torque passively aligns the satellite so that the camera points toward Earth's horizon rather than toward nadir.

The numerical simulation and experimentation presented in this work focus on ensuring stable deployment and effective gravity-gradient performance. Numerical analysis simulations were conducted to model force interactions between the satellite and the deployed mass and to compute rotational dynamics using calculated inertia properties about each principal axis. These simulations evaluate attitude response under idealized microgravity conditions. Complementary laboratory experiments were performed to characterize tether deployment behavior under scaled gravity conditions. A physical test setup was constructed in which a satellite mock-up was suspended to observe boom

deployment and spool dynamics. Multiple tests were conducted to identify spool patters that minimize unwanted reaction forces and ensure smooth deployment. The forces exerted on the system by the final chose spool pattern were included in the simulation as additional disturbance forces, labeled ( $c_{\text{deploy}}$  and  $c_{\text{final}}$ ). Upon the results from body dynamics simulations and the spooled boom deployment experiment, a flight readiness validation is achieved.

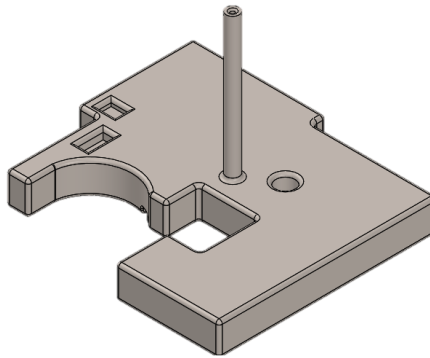
### III. Design Considerations of Boom Mass

The boom mass was designed to maximize density within the limited volume available inside the PocketQube structure. Tungsten was selected as the ideal material due to its high density of 17 to 19 g/cm<sup>3</sup>, enabling sufficient mass concentration in a compact geometry [5]. The mass is mounted along the bottom surface of the satellite baseplate to avoid interference with deployment from the launch tube and to maintain compatibility with the internal subsystem layout.

A flat plate geometry, 5 mm in thickness, was selected to conform to the external geometry of the baseplate. A cylindrical shaft extends perpendicularly from the plate toward the camera board, where the tether is anchored. A through-hole is required in both the baseplate and the EPS board to accommodate this shaft. The through-hole diameter was set to 5 mm, providing sufficient clearance for the tether to be spooled while avoiding interference with nearby printed circuit board traces and electronic components. The flat plate geometry was developed using a SolidWorks model of the baseplate to ensure that critical components, including the camera lens and light sensors, remained unobstructed.

Due to the complex baseplate layout, particular attention was given to mass distribution. To ensure smooth tether deployment, the center of mass of the boom mass was positioned as close as possible to the shaft axis, minimizing rotational inertia about the tether attachment point. The design objective was to maximize usable mass within the available footprint while maintaining rotational symmetry about the shaft. This requirement led to the incorporation of localized cutouts and contouring features that allowed the mass to surround components rather than overlap them.

Geometric refinements were performed to maintain proper center-of-mass alignment while satisfying area constraints. Minor material additions and removals were introduced to balance asymmetries imposed by internal components. To remain within PocketQube mass limitations, the final tungsten boom mass was designed to have a total mass of approximately 100 g. The resulting configuration is shown in Fig. 1.



**Fig. 1 Employed Boom Mass**

The intricate geometry of the boom mass introduces additional considerations for both simulation and experimental validation. Because the mass distribution is not perfectly symmetric about the shaft axis, the principal axes of inertia are not fully aligned with the tether axis. As the mass rotates during deployment and libration, this asymmetry may produce small variations in the effective inertia about the tether attachment point, potentially introducing higher-order dynamic effects not captured in simplified analytical models [6].

The primary objective of this study is to demonstrate nadir-pointing capability using a gravity gradient boom configuration. Accordingly, these higher-order inertial effects were not explicitly modeled in the current analysis.

However, they will require consideration in a fully flight-optimized design to ensure predictable dynamic performance. Additionally, the complex external geometry increases the risk of tether snagging or irregular deployment behavior. Experimental testing was conducted to evaluate spool patterns and deployment characteristics, and mitigation strategies were implemented to reduce the likelihood of tangling during unraveling.

#### IV. Simulation

A high-fidelity numerical simulation was developed to evaluate the coupled passive and active attitude control performance of the satellite. The simulation models the satellite and tethered gravity-gradient boom as two rigid bodies connected through a deployable, viscoelastic tether, with translation and rotational dynamics propagated simultaneously using near-circular low Earth orbit at 640 km altitude, consistent with typical PocketQube deployment altitudes from rideshare launches [7].

Gravity-gradient torque is modeled using the classical rigid body formulation, in which differential gravitational attraction across an extended body produces a restoring moment proportional to the inertia tensor asymmetry [8]. The total torque acting on each body is composed of said body's tether-induced torque and gravity-gradient torque, with a linear damping term proportional to the deviation from the local orbital angular rate is included to represent passive attitude stabilization toward the nadir-pointing equilibrium consistent with prior gravity-gradient studies showing that even minimal dissipation significantly improves pointing stability [8].

Simulation outputs include angular displacement relative to the local orbital frame, tether force components, tether-induced moments, and deployment length over time. These results confirm that the coupled dynamics produce a stable, Earth-pointing configuration suitable for the desired imaging operations.

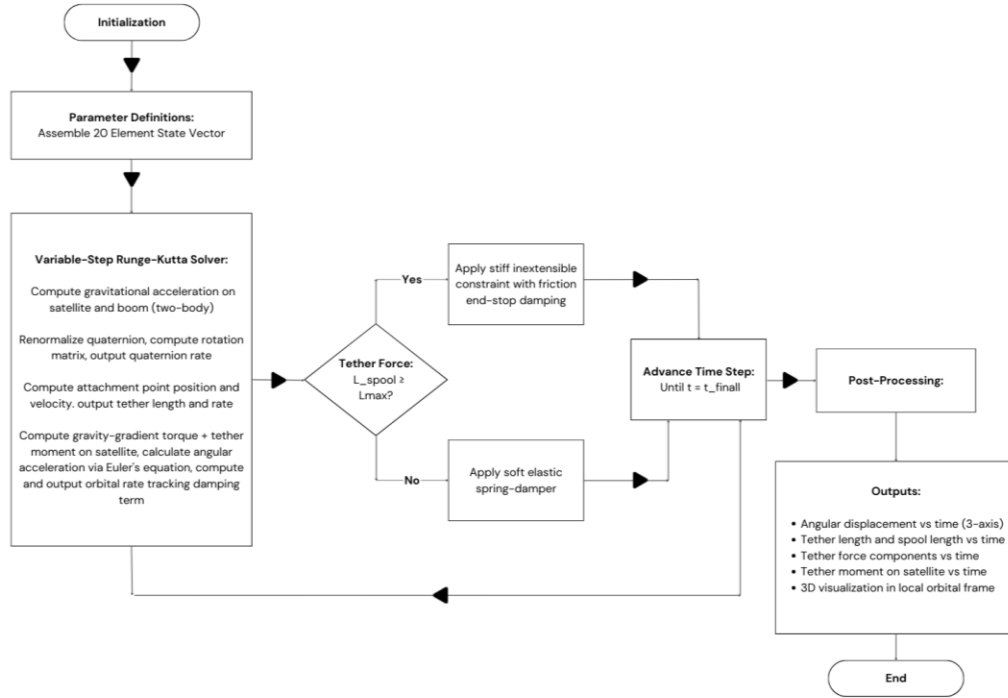


Fig. 2 Simulation Flow Chart

#### V. Physics Properties within Simulation

##### A. Theoretical Gravity Gradient

Gravity gradient stabilization is a passive attitude control technique that exploits the natural variation of Earth's gravitational field across an extended body in orbit. Because gravitational force is slightly stronger on the portion of the spacecraft closest to Earth, a differential torque is generated when the satellite principal axes are misaligned with the local vertical. This torque rotates the satellite so that its axis of maximum moment of inertia aligns with the nadir

direction [9,10]. By intentionally designing the mass distribution using a deployable, weighted boom to increase inertia asymmetry, this effect is harnessed to achieve passive nadir-pointing, providing an energy-efficient attitude control solution well suited to the mass, volume, and power constraints of the PocketQube platform [11,12].

## B. Quaternion and Euler Angles

To model the rotational dynamics of the satellite without introducing mathematical singularities, the simulation utilizes a quaternion-based attitude representation. Unlike Euler angles, which suffer from gimbal lock when two rotational axes align, quaternions provide a globally nonsingular translation of three dimensional orientation [13]. This property is especially important for gravity-gradient stabilized systems, which may experience large angular excursions during deployment before settling into equilibrium.

The quaternion state vector is propagated using the standard first-order kinematic differential equation [13]:

$$\dot{q} = \frac{1}{2} \Omega(\omega) q, \quad \#(1)$$

where  $\dot{q}$  is the quaternion rate of change,  $\Omega(\omega)$  is the quaternion rate matrix that uses  $\omega$  as the body angular velocity vector, and  $q$  is the attitude quaternion representing satellite orientation. To prevent numerical drift, the quaternion is renormalized at each integration step. To accomplish this efficiently, as well as provide detailed results in key timeframes of boom deployment, a variable-step Runge-Kutta solver is employed in the mathematical simulation. This ensures that the attitude solution remains physically meaningful throughout the simulation duration and does not slowly diverge from the real behavior of the system. The conversion of  $\omega$  into  $\Omega(\omega)$  is then used to compute transformations between the body and inertial frame states of the satellite and boom. These transformations are required for torque evaluation, tether attachment point propagation, and gravity-gradient torque calculation. The variable-step approach allows for accurate modeling of intense physical events such as the tether snapping taut at the end of deployment, as well as time-efficient computation of important values such as quaternion angles, creating an accurate and quick computational model.

## C. Motion Studies

The translational motion of both the satellite and the deployable boom are governed by the classical two-body gravitational model [14]:

$$a = -\left(\frac{\mu}{r^3}\right)r \quad \#(2)$$

By modeling the satellite and boom as independent point masses subject to identical central gravity, the simulation ensures that tether dynamics emerge naturally from relative motion rather than from imposed kinematic constraints. The deployable boom is modeled as a point mass connected to the satellite by a massless, elastic tether. This allows the entire setup to act as a spring-damper system, exerting tension only when the separation distance exceeds the commanded spool length.

The tension force is computed as:

$$T = k(L - L_{spool}) + c \dot{L} \text{ for } L > L_{spool}; T = 0 \text{ otherwise} \quad \#(3)$$

Here,  $k$  represents the elastic stiffness and  $c$  represents the damping coefficient. The damping term is applied only during extension to prevent nonphysical compressive forces. This method has been used in similar simulations before and has not been shown to affect simulation accuracy [15]. The spool length increases at a prescribed constant deployment rate until the maximum length is reached. Retraction is not modeled. This assumption reflects a simplified deployment mechanism and allows the simulation to focus on post-deployment gravity-gradient behavior. The physical errors potentially accrued from our spooling method can be addressed through physical testing, as described in section VII. Experiments. Similarly, the tether is assumed massless. While a distributed-mass tether model would introduce additional modes and longitudinal wave propagation effects [15], such complexity is beyond the scope of this first-order stability analysis. For short tether lengths and small masses typical of PocketQube platforms, this mass approximation is sufficient to capture the governing dynamics.

#### D. Local Vertical Local Horizontal (LVLH) Frame

To evaluate nadir alignment performance, spacecraft orientation is compared against LVLH reference frame. LVLH is a reference plane that remains aligned with the earth as it orbits, making it useful for checking satellite-earth alignment. The LVLH frame is constructed from the instantaneous orbital state using the following conversions:

$$\text{Radia Direction: } x_L = -\frac{r}{|r|} x_L = -r / ||r|| \quad \#(4)$$

$$\text{Along Track: } y_L = z_L \times x_L \quad \#(5)$$

$$\text{Orbit Normal: } z_L = \frac{(r \times v)}{|r \times v|} \quad \#(6)$$

This construction ensures consistency even under numerical integration error. The LVLH angular velocity is computed directly from orbital angular momentum:

$$\omega_{LVLH} = h / r^2 \quad \#(4)$$

#### E. Omitted Physical Characteristics

Several environmental perturbations were intentionally excluded: Earth oblateness (J2), atmospheric drag, solar radiation pressure, magnetic torques, tether mass distribution, and structural flexural dynamics. These omissions isolate the fundamental gravity-gradient stabilization behavior, providing a clean baseline against which passive nadir alignment can be evaluated. Most of these are commonly exclude even in higher-order models [16], though future work may incorporate J2 perturbations to assess long-term attitude drift and orbital plane precession interactions.

### VI. Simulation Results

The simulation was run over an 8000 second window, encompassing approximately 1.3 orbital periods, to capture both the full tether deployment phase and the subsequent gravity-gradient stabilization behavior. The tether deploys linearly from its initial length of 0.05 m to its full extent of 0.75 m over approximately 1556 seconds, as shown in Fig. 3, consistent with the prescribed spool rate of  $4.5 \times 10^{-4}$  m/s determined from the initial spring deployment velocity determined by the optimization of this model. Following full deployment, the actual tether length briefly overshoots  $L_{max}$  before settling at the maximum length, reflecting the kinetic energy absorption of the tether wraps acting as a friction brake on the deployment rod. After  $t \approx 1800$  s, the tether length remains stable at 0.75 m for the remainder of the simulation, confirming that the end-stop mechanism successfully arrests boom motion, and that the gravity-gradient force is sufficient to maintain tension throughout the post-deployment period.

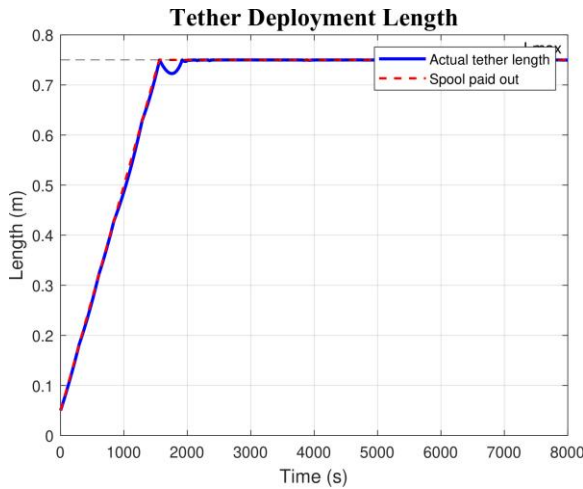


Fig. 3 Tether Deployment Length

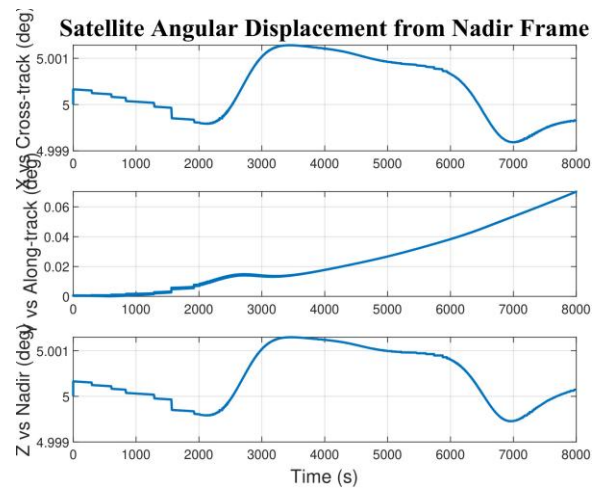
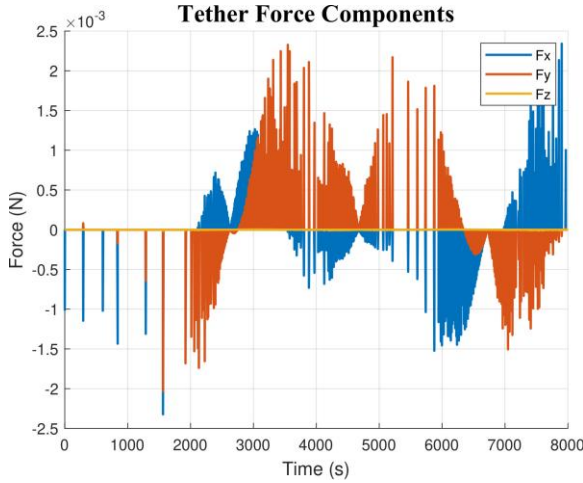
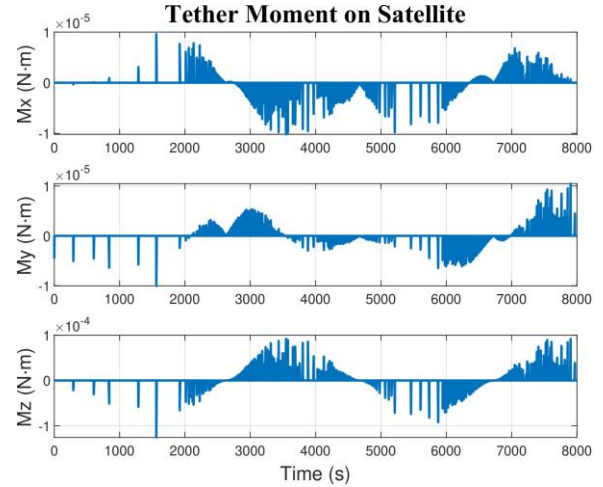


Fig. 4 Satellite Angular Displacement from Nadir Frame

The satellite angular displacement results, as shown in Fig. 4, demonstrate that nadir-pointing stability is maintained throughout the simulation. The Z vs Nadir angle, which represents the primary nadir-pointing axis, remains within a narrow band of approximately  $4.999^\circ$  to  $5.001^\circ$  for most of the simulation duration, indicating that the initial  $5^\circ$  misalignment implemented as a margin of error into the simulation would neither grow nor diverge in a real scenario. Both the X vs Cross-track and Z vs Nadir axes exhibit a smooth libration oscillation centered on the equilibrium attitude, with a period of approximately 3200–3500 seconds. This is consistent with the theoretical gravity-gradient libration period of  $T = 2\pi/\sqrt{(3n^2)} \approx 3490$  s for a circular orbit at 640 km altitude, providing strong validation of the simulation's physical accuracy. The Y vs Along-track angle exhibits a slow monotonic drift from  $0^\circ$  to approximately  $0.065^\circ$  over 8000 seconds, which is expected, since pure gravity-gradient stabilization provides no yaw restoring torque, and this residual drift would require a supplementary active control element to correct.

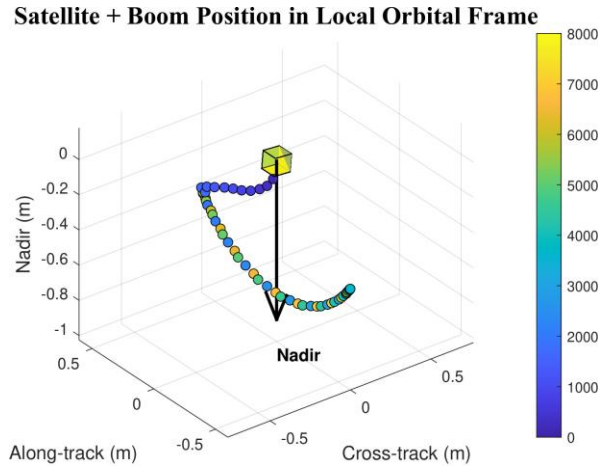


**Fig. 5 Tether Force Components (Inertial Frame)**



**Fig. 6 Tether Moment on Satellite (Body Frame)**

The tether force and moment histories, seen in Fig. 5 and Fig. 6, are consistent with the deployment and libration dynamics described above. During the deployment phase prior to  $t \approx 1556$  s, tether forces are small, on the order of  $10^{-4}$  N, reflecting the soft elastic tether model. Following full deployment,  $F_x$  and  $F_y$  grow to peak magnitudes of approximately  $2 \times 10^{-3}$  N as the boom reaches the apex of its first libration swing near  $t \approx 3200$  s, then decay as the system returns toward equilibrium. The tether moments remain on the order of  $10^{-5}$  N·m throughout, consistent with the small moment arm of the tether attachment point and the low tether forces involved. The oscillatory structure of both force and moment histories mirrors the libration period observed in the angular displacement results, further confirming internal consistency of the simulation.



**Fig. 7 Satellite plus Boom Position in Local Orbital Frame**

The three-dimensional visualization of the boom position in the local orbital frame (Fig. 7) illustrates the full deployment and libration trajectory. Early timesteps show the boom deploying directly below the satellite along the nadir axis, consistent with the guide key constraining the ejection direction. At full deployment, the boom position traces a smooth arc in the nadir-along-track plane as the libration oscillation develops, with the boom remaining below the tether attachment point throughout. No cross track divergence is observed, confirming that the tether and gravity-gradient torque together suppress out-of-plane motion effectively. The satellite body orientation, shown by the colored cubes, remains consistently nadir-pointing across all sampled timesteps, with only the small libration-driven angular variation visible. Again, these could be addressed using active attitude control stabilization mechanisms.

Overall, the simulation results confirm that the proposed gravity-gradient stabilization architecture can maintain nadir-pointing stability within approximately  $\pm 0.001^\circ$  in pitch and roll following full boom deployment. The libration oscillation, while persistent in the absence of a damper, remains bounded and small in amplitude. These results support the viability of the passive gravity-gradient approach for the intended Earth-imaging mission and establish a clear baseline against which hardware test results can be compared.

## VII. Experiment

### A. Experimental Setup and Methodology

To complement numerical simulations and evaluate the physical behavior of the gravity gradient boom system, scaled laboratory experiments were conducted to investigate tether deployment dynamics. While true microgravity conditions cannot be replicated in our standard laboratory environment, simplified ground based experiments were designed to capture first-order deployment behavior and force reactions during unspooling. Mathematic simulations were used to model idealized microgravity dynamics; however, accurately simulating tether deployment and spool behavior under realistic mechanical constraints proved challenging. Therefore, controlled drop experiments were performed under terrestrial gravity to experimentally characterize tether unspooling behavior and reaction forces.

The experimental setup consisted of a suspended satellite mock-up mounted within a basic pulley system. A tether made of Kevlar twisted cord with a diameter of 0.291 mm was routed over the pulley, attached to the satellite at one end and to a fixed support at the other. The Kevlar twisted cord was chosen for its small diameter and strong break strength of 30.0 lbf. This configuration allowed the satellite to remain suspended at a fixed height while permitting the pre-spooled boom mass to drop vertically during deployment. The tether length selected for testing was 1.95 m. Preliminary trials were conducted to determine an appropriate deployment length based on observed unspooling behavior and available internal spool volume within the satellite structure. These tests indicated that a length of 1.95 m provided sufficient deployment distance to observe steady state unspooling characteristics while remaining compatible with packaging constraints.

Due to material and fabrication constraints, the experimental boom mass could not be constructed with the same density and compact volume as the tungsten mass used in simulation. Replicating the geometry of initial design application with equivalent density would have required precision machining of tungsten, which was not feasible within the scope of laboratory testing. As a result, alternatives were investigated to produce a mass of identical shape and similar density. Consequently, a hollow, 3D printed replica of the boom mass was fabricated and filled with lead pellets, selected for their relatively high density and availability. After filling, the experimental mass was measured to be 17.7 g, significantly lower than the intended 100 g mission representative mass.

To preserve proportional inertial relationships within the experimental system, the satellite mockup mass was scaled accordingly. The experimental boom mass represents approximately 17.7% of the intended design mass, therefore, the satellite mockup was scaled to the same percentage of its nominal 150 g mass. Brass counterweights were added to the satellite structure to achieve a total mockup mass of 28.7 g. This proportional scaling maintains a consistent mass ratio between the satellite and deployed mass, allowing tether force interactions and deployment behavior to be evaluated in a controlled manner. While the absolute inertia properties differ from the flight configuration, this approach preserves the relative mass distribution necessary to characterize qualitative deployment dynamics and transient force response [17].

For each trial, the pre-spooled mass was released from an approximate height of 2 m relative to the suspended satellite model, while remaining attached to the camera board. Multiple deployment tests were conducted to evaluate

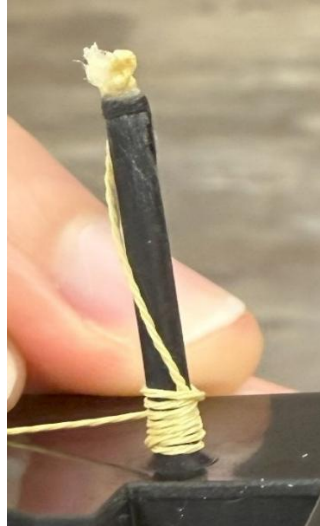
different spool patterns and their influence on unspooling smoothness and reaction forces. Initial configurations produced intermittent force spikes that caused abrupt changes in tether tension, which in an orbital environment could introduce unwanted attitude disturbances. Subsequent refinements to spool geometry demonstrated that steady-state unspooling behavior was achievable, minimizing transient force spikes during most of the deployment. These results inform the selection of an optimal spool pattern for further validation and simulation comparison.

## B. Deployment Experiments

Four spool configurations were evaluated to characterize deployment smoothness and transient tension behavior. The primary performance metric was the presence and magnitude of transient tension spikes during unspooling, as these directly correlate to potential attitude disturbances in an orbital environment.



**Fig. 8 Alternating Up-Down Spooling**



**Fig. 9 Single Direction Bottom-to-Top Spooling**



**Fig. 10 Transitional Damping Between Layers**

### 1. *Configuration 1: Preliminary Alternating Up Down Spooling*

The first spool configuration consisted of alternating layers wound clockwise from top to bottom and then bottom to top along the shaft, resulting in six total layers (Fig. 8). This configuration produced significant transient tension spikes during the deployment. When transitioning between upward and downward spool layers, abrupt changes in the tether exit direction induced impulsive forces in the mass, resulting in rapid changes in tether tension and observable satellite disturbance. Additionally, when unspooling occurred in the direction that receded up the shaft, previously wound layers introduced frictional interference, reducing deployment momentum and contributing to irregular unspooling.

### 2. *Configuration 2: Full System Validation of Configuration 1*

The second experiment applied the same spool pattern to the complete suspended satellite system. Results were consistent with preliminary observations, confirming that the alternating up-down configuration generated persistent tension spikes and irregular deployment behavior. However, near the end of deployment, when unspooling occurred predominantly in the direction from top to bottom of the shaft, the system exhibited brief periods of steady-state behavior. This observation informs subsequent spool modifications.

### 3. *Configuration 3: Single Direction Bottom-to-Top Spooling*

To reduce transition-induced disturbances, the third configuration eliminated alternating directional layers. The tether was spooled only from bottom to top, with the tether routed as straight as possible along the shaft between layers (Fig. 9). This configuration significantly reduced tension spikes and produced more consistent unspooling behavior. Minor transient disturbances were still observed during inter-layer transitions, but overall deployment smoothness improved substantially compared to previous configurations.

#### 4. *Configuration 4: Transitional Damping Between Layers*

The fourth configuration introduced controlled transitional winding between layers to reduce abrupt direction changes. By incorporating limited intermediate wrapping during downward transitions (Fig. 10), small frictional resistance was intentionally introduced to smooth inter-layer transitions. This configuration resulted in near-continuous steady-state unspooling for most deployment. Remaining interruptions were attributed primarily to anchor geometry rather than spool configuration.

### C. Conclusion of Experiment Results

The third and fourth configurations demonstrated that steady-state tether deployment is achievable through appropriate spool geometry. Among the configurations tested, the refined transitional winding pattern provided the most stable deployment with minimal transient tension spikes. While additional optimization is required for flight implementation under microgravity conditions, terrestrial testing confirms that spool geometry plays a dominant role in mitigation of deployment-induced attitude disturbances.

## VIII. Discussion

The simulation and physical testing results provide mutual validation of the proposed deployment mechanism. The numerical model confirmed that transient tensions spikes during deployment produces attitude disturbances scaling proportionally with peak tether force, while physical testing independently identified spool geometry as the dominant factor governing tension spike amplitude. The libration oscillation observed in the simulation, characterized by a period of approximately 3490 seconds, is an expected and bounded consequence of passive gravity-gradient stabilization rather than an indication of instability; a behavior well-established across missions from the earliest gravity-gradient experiments in the 1960s through modern nanosatellite designs [11]. The slow yaw drift of approximately 0.065 degrees over the 8000-second simulation window is not controlled by the passive gravity-gradient mechanism, though this is not a practical concern given the planned implementation of magnetorquers, which will actively stabilize divergence in yaw [18].

A fundamental limitation of the experimental analysis is that all spool characterization and deployment trials were conducted under Earth's gravitational field, introducing loading conditions absent in the orbital environment. Ground testing improves gravitational preloading on tether wraps that increase friction and affects deployment rate in ways that microgravity would not replicate, meaning the refined winding pattern's behavior cannot be fully characterized without parabolic flight or on-orbit demonstration. Importantly, however, the factors introduced by terrestrial testing act conservatively – the additional forces present on the ground would only serve to further stabilize deployment – meaning the methods presented here can be expected to remain in effective in orbit.

Some development work remains before the system can be considered flight ready. On the simulation side, the addition of a magnetic torque model would enable direct assessment of the magnetorquer yaw control strategy intended for use with this satellite. On the hardware side, integration of an attitude determination capability, whether through magnetometers or a commercial IMU, would allow simulation predictions to be validated against in-flight telemetry following deployment. Nevertheless, the results presented in this study establish a credible foundation for the passive gravity-gradient approach, and the remaining work is well defined and achievable within constraints of the PocketQube platform.

## IX. Conclusion

The design and validation of a Gravity Gradient boom arm for the Wentworth PocketQube is achieved using a microgravity dynamics simulation in tandem with experimental setups replicating tether dynamics. The information yielded from microgravitational dynamic simulations provides critical understanding of expected low-earth orbit behavior of the Gravity Gradient system. The intimate understanding of failure modes, simulation results, and mitigation techniques informs design choices to ensure a robust, flight ready system. For example, satellite-mass inversion can be mitigated upon consideration of the simulation results. Failure modes such as tether knotting, kinking, or catching are of concern, and stem from convolutions regarding spooling pattern. Due to the complexity of this failure mode, an experimental approach was taken as opposed to simulating this effect. Qualitatively analyzing the unspooling behavior of the boom arm system over iterations of spooling patterns informs a final spooling pattern decision for flight. Given experimental and simulation results, a final flight-ready design is properly informed and built.

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